

# FoodEnergyMAS — Security Assessment Report

Session: `139a70d7-ad56-40bc-9cc0-d583e71777a8` Generated: 2026-06-16 16:32 UTC  
Scenario: Regional drought assessment — North Africa grain belt, Q3 2026

## Executive Summary

*Activate emergency grain reserves for Akmola region. Negotiate duty-free wheat import from Russia (300k tonnes). Subsidise electricity for irrigation by 30%.*

| Confidence | Escalation                                                                           | Import needed | Reserves release |
|------------|--------------------------------------------------------------------------------------|---------------|------------------|
| 78%        |  No | YES           | YES              |

## Agent Findings

 Farmer Agent —  HIGH

- Climate risk score: 0.72
- Max regional deficit: 22.0%
- Deficit regions: Akmola, Kostanay
- Risk flags: drought\_risk, fertilizer\_shortage

Crop Forecast:

| Crop  | Region     | Expected (t) | vs Last Year |
|-------|------------|--------------|--------------|
| wheat | Kazakhstan | 17,800,000   | +7.0%        |

 Logistics Agent —  MEDIUM

- Supply chain loss estimate: 8.5%
- Available storage: 2,500,000 tonnes
- Bottlenecks: Dostyk border crossing, Aktau port

Route Plan:

| From     | To     | Mode | Capacity (t) | Days | Cost/t |
|----------|--------|------|--------------|------|--------|
| Kostanay | Akmola | rail | 50,000       | 3    | \$12.5 |

⚡ Energy Agent — 🟡 MEDIUM

- Price: \$0.18/kWh (rising)
- Grid load: 82.0%
- Shortage alert: ⚠️ WARNING
- Renewable share: 22.0%
- Impact on food production: High electricity costs increase irrigation and cold storage expenses by 25%.

📈 Market Agent — 🟡 MEDIUM

- Demand signal: high (increasing)
- Stock level: 28.0 days
- Inflation risk: medium
- Speculation detected: No

Price Recommendations:

| Commodity | Current | Recommended | Rationale                                                       |
|-----------|---------|-------------|-----------------------------------------------------------------|
| wheat     | \$195.0 | \$210.0     | Supply deficit warrants 8% price increase to reduce speculat... |

Policy Decision

Recommendation: Activate emergency grain reserves for Akmola region. Negotiate duty-free wheat import from Russia (300k tonnes). Subsidise electricity for irrigation by 30%. Subsidy Plan:

| Target                 | Amount (USD M) | Duration | Rationale                                                  |
|------------------------|----------------|----------|------------------------------------------------------------|
| irrigation_electricity | \$45.0M        | 6 months | Offset 40% energy cost increase to protect harvest yields. |

Emergency Import: Emergency import of 300,000 tonnes of wheat from Russia via Dostyk corridor.

Audit Trail

- Debate rounds: 0
- Total actions logged: 5

• Final decision ready: True

| # | Agent     | Action    | Details             |
|---|-----------|-----------|---------------------|
| 1 | farmer    | published | payload_id=9c9aa5b6 |
| 2 | energy    | published | payload_id=988d490d |
| 3 | logistics | published | payload_id=cee6ac43 |
| 4 | market    | published | payload_id=e41da85a |
| 5 | regulator | published | payload_id=c7402136 |

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